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The University of Lahore
Department of Computer Science & IT
Midterm Examination



Fall-2024

Course Title:	Theory of Automata	Course Code:	CS09305	Credit Hours:	3
Course Instructors:	Dr. Toqir Rana, Dr. Shahid Yousaf, Mr. Naveed Ahmad Khan, Mr. Muhammad Tayyab, Ms. Iqra	Program Name:	BSCS		
Time Allowed:	90 Minutes	Maximum Marks:	50		
Date:	26-11-24	Course Mentor:	Dr. Toqir Rana		
Student's Name:	Maham Abdul Sattar	Signature:			
Reg. No:	70141633	Section:	F		

Question # 1: (10 marks)

Give regular expressions for the following languages where $\Sigma = \{a, b\}$

- Any string that does not start with double letter i.e. aa or bb.
- $\{w \mid \text{every odd position of } w \text{ is } a\}$

Question # 2: (3x5 =15 marks)

Construct the DFA for each of the following language where $\Sigma = \{0, 1\}$

- $\{w \mid w \text{ is any string when } 0 \text{ comes it should be the multiple of } 3\}$
- $\{w \mid w \text{ doesn't contain the substring } 110\}$
- $\{w \mid w \text{ is an empty set}\}$

Question # 3: (6 marks)

Construct the NFA for each of the following language where $\Sigma = \{x, y\}$

- The language $y^* (xxy^*)^*$ with three states
- $\{w \mid w \text{ contains an even number of } x, \text{ or contains exactly two } y\}$

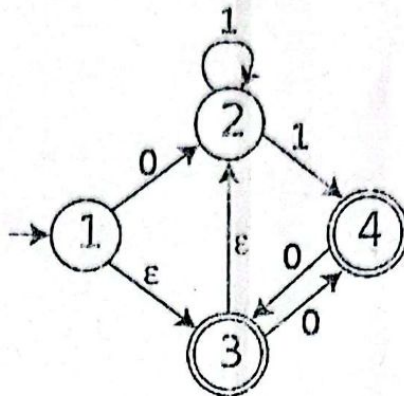
Question # 4: (4 Marks)

Construct a TG for the language that start and end with different letter and where b always comes on the second position, over $\Sigma = \{a, b\}$

Question # 5:

(10+3+15 Marks)

a. Convert the following NFA to its equivalent DFA



b. Convert the following DFA into the RE using state elimination method.

